Supplement 5. eAppendix

R script for the primary outcome and a sensitivity analysis (mortality at the longest follow up)

```
## ------------------------------------------------------------------------------------------------------------------
## Packages

library(readr)
library(meta)
library(netmeta)




## ------------------------------------------------------------------------------------------------------------------
# Primary outcome: All cause mortality at the longest follow-up wihtin 1 year but 90 days or longer

## Format trial data

vitc_pair1 <- pairwise(
  treat = treatment,
  event = r,
  n = n,
  sm = "OR",
  data = vitc_outcome1,
  studlab = study
)

vitc_pair1


## ------------------------------------------------------------------------------------------------------------------
## Conduct random effects network meta-analysis

trts <- c("placebo/usual care","HD-vitC+GC+vitB1", "HD-vitC+vitB1", "GC")

net.out1 <- netmeta(TE, seTE, treat1, treat2, studlab, data = vitc_pair1, sm = "OR", seq = trts, comb.fixed = FALSE, prediction = TRUE)
net.out1




## ------------------------------------------------------------------------------------------------------------------
## funnel plot
```

```
png(file="funnel.out1.png")

ord <- c("HD-vitC+GC+vitB1", "HD-vitC+vitB1", "GC", "placebo/usual care")

funnel(net.out1,
    order = ord)

dev.off()


## ---------------------------------------------------------------------------------------------------------------
## p-scores

nr.outcome1 <- netrank(net.out1)
print(nr.outcome1)



## ---------------------------------------------------------------------------------------------------------------
png(file="forest.out1.png", width = 800, height = 800)

split.out1 <- netsplit(net.out1, upper = FALSE, reference.group = "placebo/usual care",
baseline.reference = TRUE)
forest.netsplit(split.out1, show ="all")

dev.off()

## side split
print(split.out1, all = FALSE, digits = 2)


## ---------------------------------------------------------------------------------------------------------------
## forest plot for a pair-wise comparison

pair1.GC.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair1, subset = treat1 == "GC", studlab = studlab, sm = "OR")
summary(pair1.GC.PBO.OR)

png(file = "pair1.png", width = 800, height = 800)

forest(pair1.GC.PBO.OR,
    sortvar = seTE,
    allstudies = F,
    print.tau2 = TRUE,
    print.tau2.ci = TRUE,
    comb.fixed = FALSE,
    fs.study = 10,
```

```
    ff.study = "italic",
    col.diamond = "red",
    col.diamond.lines = "red",
    col.square = "blue",
    col.square.lines = "blue")

dev.off()


## ---------------------------------------------------------------------------------------------------------------
## design-by-treatment

net.out1$d

designs.out1 <- as.character(decomp.design(net.out1)$Q.het.design$design)
designs.out1


decomp.design(net.out1)


## ---------------------------------------------------------------------------------------------------------------
## Conduct component network meta-analysis

net.comb.out1 <- netcomb(net.out1,
                 inactive = "placebo/usual care",
                 sep.comps = "+",
                 comb.fixed = FALSE,
                 tau.preset = NULL)
net.comb.out1




## ---------------------------------------------------------------------------------------------------------------
# Sensitivity analysis: 2. Mortality at the longest follow up (outcome2)

## Format trial data

vitc_pair2 <- pairwise(
  treat = treatment,
  event = r,
  n = n,
  sm = "OR",
  data = vitc_outcome2,
  studlab = study
```

```
)

vitc_pair2


## ------------------------------------------------------------------------------------------------------------------
## Conduct random effects network meta-analysis

trts <- c("placebo/usual care","HD-vitC+GC+vitB1", "HD-vitC+vitB1", "VHD-vitC", "HD-vitC", "vitC", "GC","vitB1")
net.out2 <- netmeta(TE, seTE, treat1, treat2, studlab, data = vitc_pair2, sm = "OR", comb.fixed = FALSE, seq = trts,  prediction = TRUE)

net.out2




## ------------------------------------------------------------------------------------------------------------------
## funnel plot

png(file="funnel.out2.png")

trts <- c("HD-vitC+GC+vitB1", "HD-vitC+vitB1", "VHD-vitC", "HD-vitC", "vitC", "GC","vitB1", "placebo/usual care")

funnel(net.out2, order = trts, legend = FALSE)

dev.off()

png(file="funnel.out2.legend.png")

trts <- c("HD-vitC+GC+vitB1", "HD-vitC+vitB1", "VHD-vitC", "HD-vitC", "vitC", "GC","vitB1", "placebo/usual care")

funnel(net.out2, order = trts)

dev.off()


## ------------------------------------------------------------------------------------------------------------------
## network graph

size.outcome2 <- tapply(vitc_outcome2$n, vitc_outcome2$treatment, sum)

## ------------------------------------------------------------------------------------------------------------------
png(file="netgraph_outcome2.png")
```

```
netgraph(net.out2, plastic = FALSE, thickness = "number.of.studies", points = TRUE, cex.points = 0.2*sqrt(size.outcome2), col.points = 4, col = 1, offset = 0.08, multiarm = FALSE)

dev.off()


## ---------------------------------------------------------------------------------------------------------------
## p-scores

nr.outcome2 <- netrank(net.out2)
print(nr.outcome2)



## ---------------------------------------------------------------------------------------------------------------
png(file="forest.out2.png", width = 600, height = 3000)

split.out2 <- netsplit(net.out2, sep.trts = "vs", reference.group = "placebo/usual care", baseline.reference = TRUE)
forest.netsplit(split.out2, show = "all")

dev.off()

## side split
print(split.out2, all = FALSE, digits = 2)


## ---------------------------------------------------------------------------------------------------------------
#forest plots for each pair-wise comparison

pair2.GC.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "GC", studlab = studlab, sm = "OR")
summary(pair2.GC.PBO.OR)

png(file = "pair2.GC.PBO.png", width = 800, height = 800)

forest(pair2.GC.PBO.OR,
     sortvar = seTE,
     allstudies = F,
     print.tau2 = TRUE,
     print.tau2.ci = TRUE,
     comb.fixed = FALSE,
     fs.study = 10,
     ff.study = "italic",
     col.diamond = "red",
     col.diamond.lines = "red",
     col.square = "blue",
     col.square.lines = "blue")
```

```
dev.off()

pair2.hdc.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "HD-vitC" & treat2 == "placebo/usual care",
studlab = studlab, sm = "OR")
summary(pair2.hdc.PBO.OR)

png(file = "pair2.hdc.PBO.png", width = 800, height = 800)

forest(pair2.hdc.PBO.OR,
     sortvar = seTE,
     allstudies = F,
     print.tau2 = TRUE,
     print.tau2.ci = TRUE,
     comb.fixed = FALSE,
     fs.study = 10,
     ff.study = "italic",
     col.diamond = "red",
     col.diamond.lines = "red",
     col.square = "blue",
     col.square.lines = "blue")
dev.off()

pair2.triple.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "HD-vitC+GC+vitB1" & treat2 == "placebo/usual
care", studlab = studlab, sm = "OR")
summary(pair2.triple.PBO.OR)

png(file = "pair2.triple.PBO.png", width = 800, height = 800)

forest(pair2.triple.PBO.OR,
     sortvar = seTE,
     allstudies = F,
     print.tau2 = TRUE,
     print.tau2.ci = TRUE,
     comb.fixed = FALSE,
     fs.study = 10,
     ff.study = "italic",
     col.diamond = "red",
     col.diamond.lines = "red",
     col.square = "blue",
     col.square.lines = "blue")
dev.off()

pair2.triple.GC.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "HD-vitC+GC+vitB1" & treat2 == "GC", studlab =
studlab, sm = "OR")
summary(pair2.triple.GC.OR)
```

```
png(file = "pair2.triple.GC.png", width = 800, height = 800)

forest(pair2.triple.GC.OR,
     sortvar = seTE,
     allstudies = F,
     print.tau2 = TRUE,
     print.tau2.ci = TRUE,
     comb.fixed = FALSE,
     fs.study = 10,
     ff.study = "italic",
     col.diamond = "red",
     col.diamond.lines = "red",
     col.square = "blue",
     col.square.lines = "blue")
dev.off()

pair2.vhdc.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "VHD-vitC" & treat2 == "placebo/usual care",
studlab = studlab, sm = "OR")
summary(pair2.vhdc.PBO.OR)

png(file = "pair2.vhdc.PBO.png", width = 800, height = 800)

forest(pair2.vhdc.PBO.OR,
     sortvar = seTE,
     allstudies = F,
     print.tau2 = TRUE,
     print.tau2.ci = TRUE,
     comb.fixed = FALSE,
     fs.study = 10,
     ff.study = "italic",
     col.diamond = "red",
     col.diamond.lines = "red",
     col.square = "blue",
     col.square.lines = "blue")
dev.off()

pair2.vitb.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "vitB1" & treat2 == "placebo/usual care", studlab
= studlab, sm = "OR")
summary(pair2.vitb.PBO.OR)

png(file = "pair2.vitb.PBO.png", width = 800, height = 800)

forest(pair2.vitb.PBO.OR,
     sortvar = seTE,
     allstudies = F,
```

```
    print.tau2 = TRUE,
    print.tau2.ci = TRUE,
    comb.fixed = FALSE,
    fs.study = 10,
    ff.study = "italic",
    col.diamond = "red",
    col.diamond.lines = "red",
    col.square = "blue",
    col.square.lines = "blue")
dev.off()

pair2.vitc.PBO.OR <- metabin(event1, n1,event2, n2,
            data = vitc_pair2, subset = treat1 == "vitC" & treat2 == "placebo/usual care", studlab
= studlab, sm = "OR")
summary(pair2.vitc.PBO.OR)

png(file = "pair2.vitc.PBO.png", width = 800, height = 800)

forest(pair2.vitc.PBO.OR,
    sortvar = seTE,
    allstudies = F,
    print.tau2 = TRUE,
    print.tau2.ci = TRUE,
    comb.fixed = FALSE,
    fs.study = 10,
    ff.study = "italic",
    col.diamond = "red",
    col.diamond.lines = "red",
    col.square = "blue",
    col.square.lines = "blue")
dev.off()



## ---------------------------------------------------------------------------------------------------------------
## design-by-treatment

net.out2$d

designs.out2 <- as.character(decomp.design(net.out2)$Q.het.design$design)
designs.out2

decomp.design(net.out2)


## ---------------------------------------------------------------------------------------------------------------
## Conduct component network meta-analysis
```

```
net.comb.out2 <- netcomb(net.out2,
                         inactive = "placebo/usual care",
                         sep.comps = "+",
                         comb.fixed = FALSE,
                         tau.preset = NULL)
net.comb.out2
```

[END]

Dataset is available on request, accompanied by an appropriate scientific rationale and research plan.